

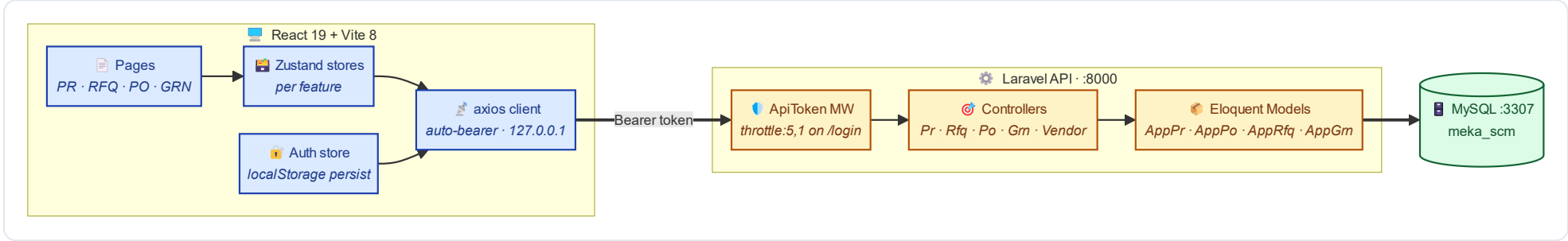
SCM — Flow Diagrams

Visual reference for how the Meka SCM system fits together. Diagrams use **Mermaid** — render in GitHub, Notion, or VS Code (with the Mermaid extension).

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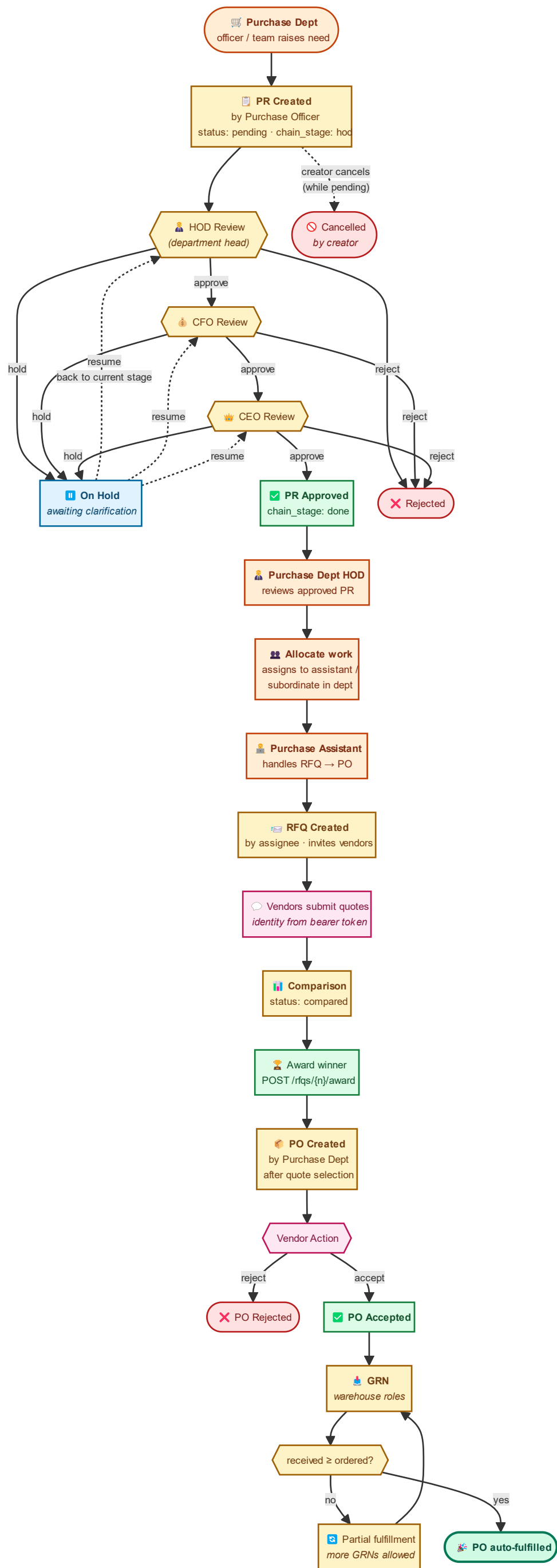
- 1. **System Architecture** — how the pieces talk to each other
- 2. **Procurement Pipeline** — PR → RFQ → PO → GRN end-to-end
- 3. **Award Approval Flow** — three-party consensus + CFO/CEO chain
- 4. **Role-Based Access** — who lands where, who can do what
- 5. **Auth & Request Lifecycle** — login, bearer flow, self-healing
- 6. **Data Model (ERD)** — core tables and relationships

1. System Architecture



2. Procurement Pipeline

End-to-end procure-to-receive flow. PRs originate in the **Purchase department** (officer / team / dept). After CEO approval, the **Purchase HOD** allocates the post-approval work (RFQ + PO authoring) to a subordinate. Every status mutation is guarded by a **terminal-state check** (409 if record is already in a terminal state).



 Access rules for this pipeline

- **RFQ create / delete** → **Purchase department only**. All other roles get read-only views (List + Detail + Comparison leaderboard).
- **Quotation comparison leaderboard** → visible to **everyone** (transparency by design).
- **Award button** → gated by the three-party consensus + CFO/CEO chain detailed in **§3 Award Approval Flow**. Only the **Purchase HOD** ultimately fires the award.

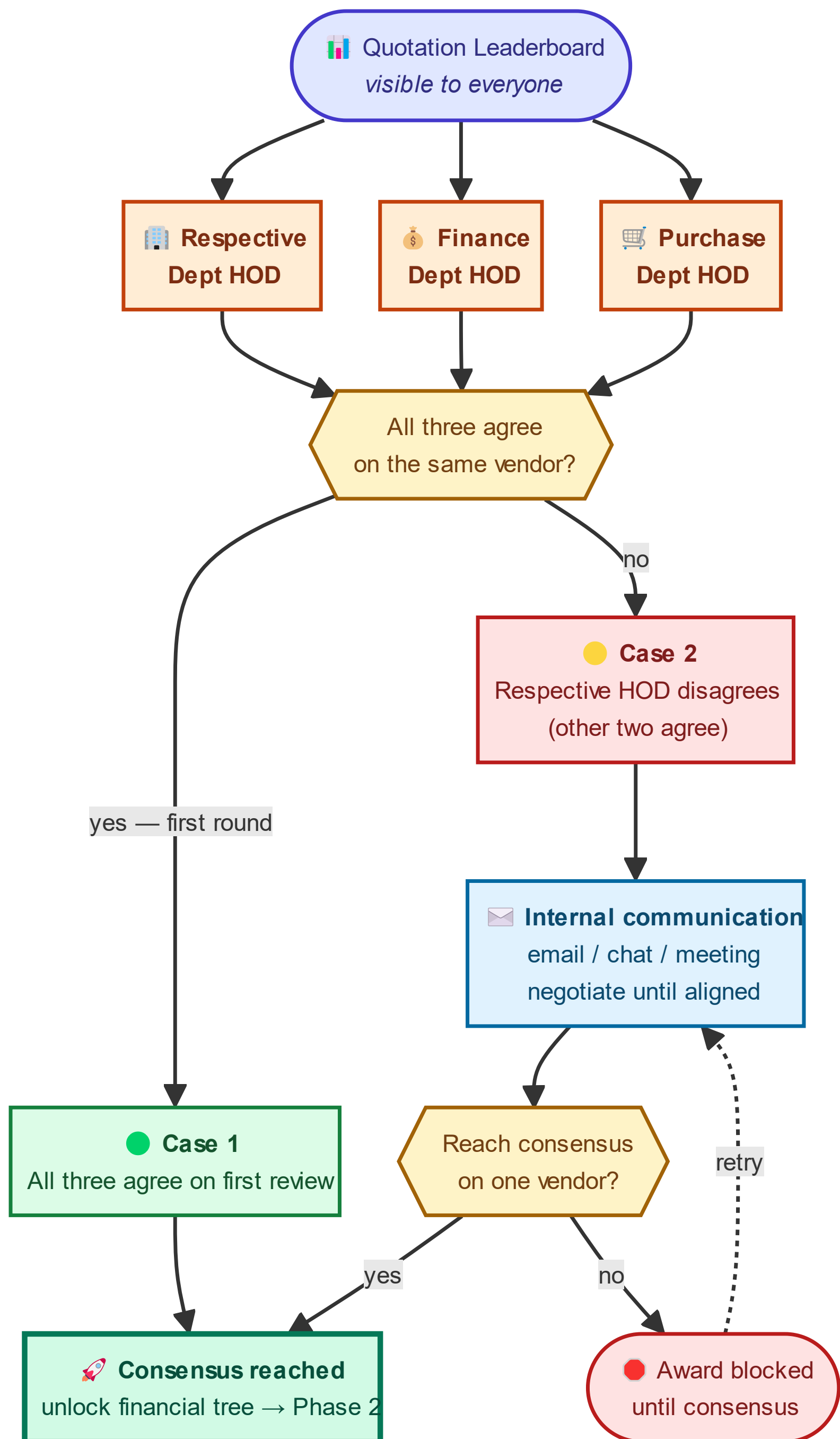
3. Award Approval Flow

The "Award" action on a quotation is **not** a single click — it requires a three-party internal consensus, then the same CFO → CEO approval tree used for PRs. Two scenarios cover whether the Respective Department HOD agrees on the first round.

Phase 1 — Three-party consensus (departmental)

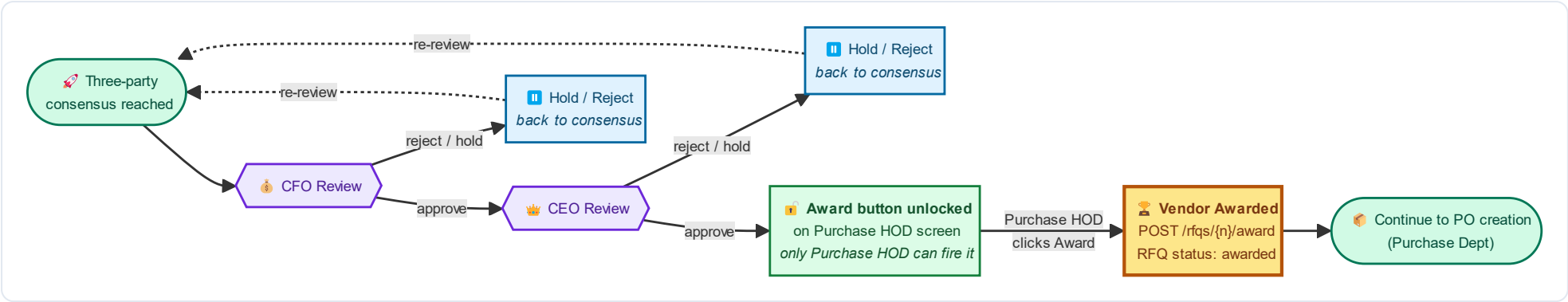
Three stakeholders must converge on **one vendor** before the financial-approval tree opens:

Party	Role	Why they review
 Respective Dept HOD	Originating department head	Confirms vendor meets the technical / operational need that drove the PR
 Finance Dept HOD	Finance head	Validates pricing, payment terms, financial fit
 Purchase Dept HOD	Purchase head	Confirms vendor compliance, history, procurement-policy fit



Phase 2 — Financial approval tree + final award

Once the three departmental HODs agree on a vendor, the decision flows up the same approval tree used for PRs. The **Award button** only becomes functional on the Purchase HOD's screen after CEO approval.

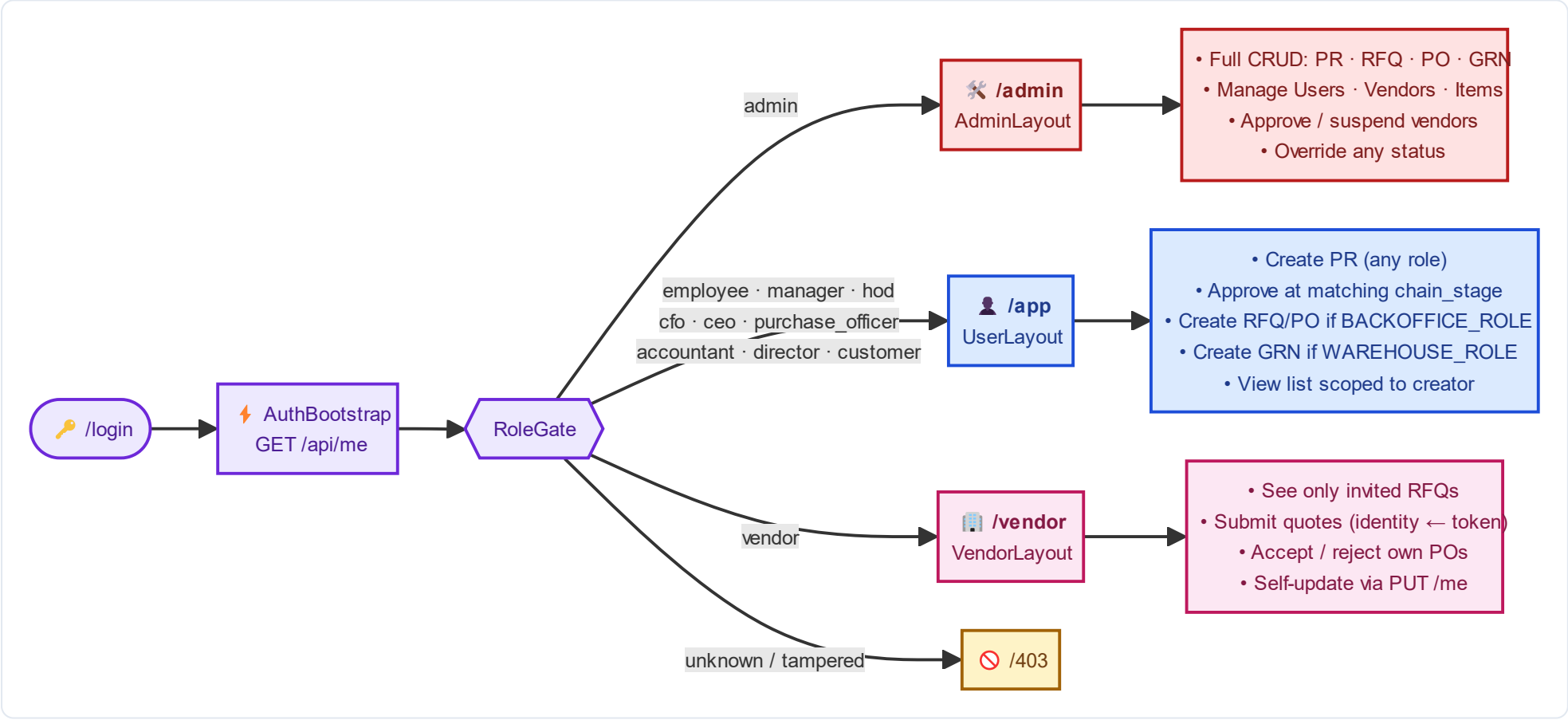


Award button visibility matrix

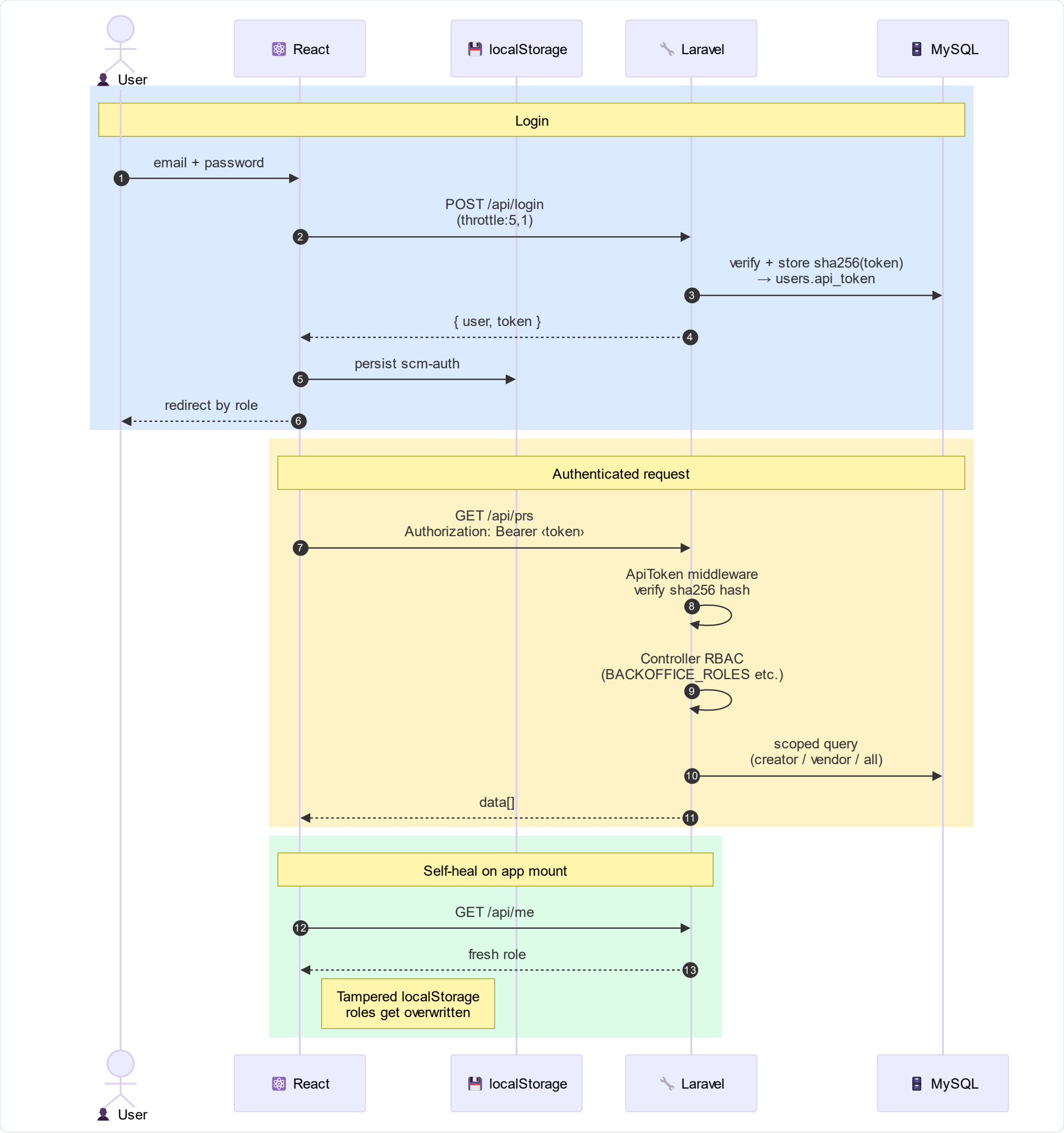
Stage	Award button visible?	Functional?	Who can fire?
Three-party consensus pending	✗ Hidden	—	—
Consensus reached, awaiting CFO	✗ Hidden	—	—
CFO approved, awaiting CEO	✗ Hidden	—	—
CEO approved	✓ Visible	✓ Yes	Purchase HOD only
Awarded (terminal)	—	—	—

4. Role-Based Access

Three audiences, three layouts, eleven roles. RBAC is enforced **defense-in-depth** — `<RoleGate>` on the route, `axios bearer in transit`, `ApiToken` middleware at the edge, controller `BACKOFFICE_ROLES` / `WAREHOUSE_ROLES` constants at the action.

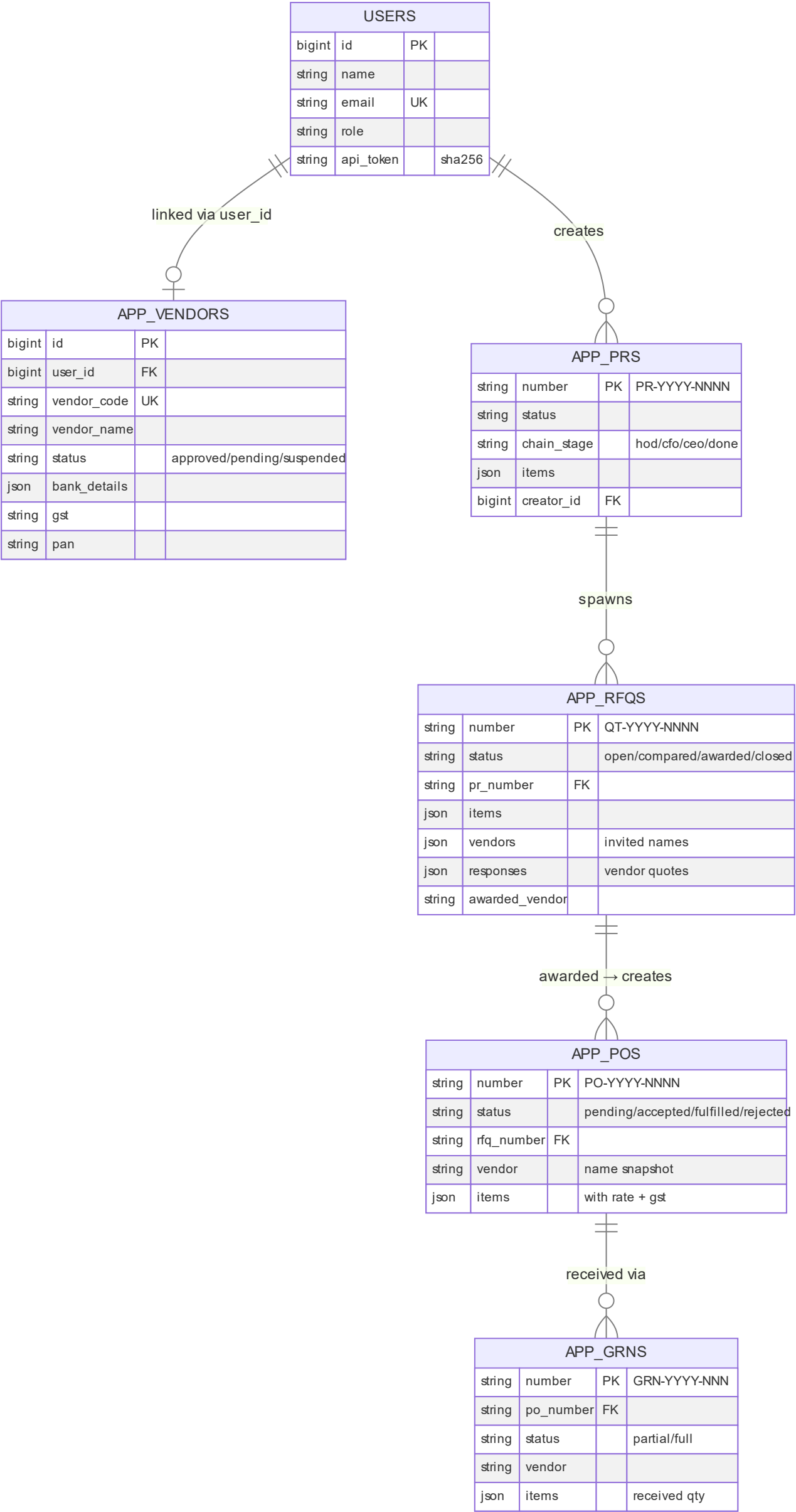


5. Auth & Request Lifecycle



6. Data Model (ERD)

Core tables. All item lists, vendor invites, and quote responses are stored as **JSON columns** on the parent record — denormalised by design for snapshot integrity.



Appendix · Auto-generated number formats

Document	Pattern	Example
PR	PR-YYYY-NNNN	PR-2026-0042
RFQ	QT-YYYY-NNNN	QT-2026-0017
PO	PO-YYYY-NNNN	PO-2026-0009
GRN	GRN-YYYY-NNN	GRN-2026-031

Appendix · Terminal states (mutation blocked, returns 409)

Entity	Terminal states
PR	approved · rejected · cancelled (hold is non-terminal — resumable)
RFQ	awarded · closed
PO	fulfilled · rejected
GRN	(no terminal — append-only)